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Education

2012 – 2021 **Doctor of Philosophy (Ph.D, Electrical Engineering),**
Hamdard University
PCD: ***26648***
Thesis title: *Efficient Methodology for Magnetic Resonance Image Reconstruction.*
Supervisor: Prof. Tariq Javid
Co-Supervisor: Prof. Arshad Aziz

2006 – 2008 **Masters in Science (MS), Electronics (Communication and Signal Processing),**
Usman Institute of Technology (Hamdard University).

2000 – 2003 **Bachelor of Science (Biomedical Engineering)**
Sir Syed University of Engineering and Technology (SSUET).

Latest Certification and Award

Feb – 2026 **The Best University Teacher Award for 2025**
Hamdard University

June – 2025 **Patent Law and Global Public Health "PatentX"**
WIPO Academy **Harvard Law School, USA**

Interests

Artificial Intelligence	AI in Medical Imaging and Communications.
Biomedical Instrumentation	ECG, EMG, EEG.
Magnetic Resonance Imaging	Compressed Sensing to reconstruct fast MR images.

Employment History

December 2025 – now **Associate Professor**,
Department of Biomedical Engineering, FEST, Hamdard University, Karachi, Pakistan.

October 2012 – Dec 2025 **Assistant Professor**,
Department of Biomedical Engineering, FEST, Hamdard University, Karachi, Pakistan.

January 2012 – October 2012 **Lecturer**,
Department of Biomedical Engineering, FEST, Hamdard University, Karachi, Pakistan.

March 2008 – January 2012 **Design and Development Engineer (Biomedical)**,
Pakistan Council of Scientific and Industrial Research (PCSIR), Karachi, Pakistan.

November 2005 – March 2008 **Service and Development Engineer**,
Electro Dynamic Works (PUMA Electronics), Karachi, Pakistan.

March 2005 – September 2005 **Trainee Engineer: Testing and Installation**,
Oasis Medical Systems, Karachi, Pakistan.

May 2004 – January 2005 **Service and Development Engineer**,
Standard Suppliers, Karachi, Pakistan.

June 2003 – July 2003 **Internee:- Biomedical Engineering Department**,
Siemens Pakistan, Karachi, Pakistan.

June 2002 – July 2002 **Internee:- Biomedical Engineering Department**,
The Kidney Centre, Karachi, Pakistan.

Publications

Journal Published Articles

15. Syed Ibad Hasnain, **Muhammad Faris**,. (2026). Deep learning in lung cancer imaging: A narrative review of CT, MRI, PET, and X-ray-based diagnostic models. Imaging, 1-10 (HEC X-category).
https://doi.org/10.1556/1647.2026.00454

14. Hafiza Syeda Yusra Tirmizi, Syed Ibad Hasnain, **Muhammad Faris**, Rabail Khowaja, Saad Abdullah. (2026). Predictive Modelling of Natural Medicinal Compounds for Alzheimer disease Using Machine Learning and Cheminformatics. arXiv preprint arXiv:2604.18316.
https://doi.org/10.48550/arXiv.2604.18316

13. Syed Ibad Hasnain, **Muhammad Faris**, Hafiza Syeda Yusra Tirmizi, Rabail Khowaja, Hafsa Israr (2026). CNN-ViT Fusion with Adaptive Attention Gate for Brain Tumor MRI Classification: A Hybrid Deep Learning Model. arXiv preprint arXiv:2604.23137.
https://doi.org/10.48550/arXiv.2604.23137

12. **Muhammad Faris,** Falak Khalid, Maida Shahid, ul Haque Zeeshan, S.I Hasnain, Ali Mir,Farooq. (2026). An Ensemble Deep Learning Framework for Automated MultiClass Skin Lesion Classification Using ConvNeXt-Tiny, EfficientNet V2-S, and Mobile NetV3. VFAST Transactions on Software Engineering, 60–72 (HEC Y-category).
https://doi.org/10.21015/vtse.v14i1.2311

11. Syed Ibad Hasnain, **Muhammad Faris,** Hafiza Syeda Yusra Tirmizi, Rabika Kamal, Tariq Javid. (2025). Simulation-Based Machine Learning Approach to Classify Accelerated Biological Aging Using Telomere and Epigenetic Markers. Sir Syed University Research Journal of Engineering & Technology, 1-9 (HEC Y-category).
https://doi.org/10.33317/ssurj.700

10. **Muhammad Faris**, Qayyum Ramsha, Ali Mir,Farooq, Mughal Muhammad,Mansoor, Javid Tariq. (2025). MobileNetV2 and EfficientNetB3 Ensemble for Scalable Dermoscopic Skin Lesion Classification. *Allied Medical Research Journal*, 97–104 (HEC Y-category).
https://doi.org/10.59564/amrj/03.04/014

9. **Muhammad Faris**, Shahzaib Khalid, S.I Hasnain, Musfira Jamil, Huma Yasmeen, Mariam Arif. (2025). Scalable Hybrid Deep Learning-Based Architecture for Glaucomatous and Healthy Eye Classification in Retinal Fundus Images. *VFAST Transactions on Software Engineering*, 1–12 (HEC Y-category).
https://doi.org/10.21015/vtse.v13i4.2232

8. Syed Ibad Hasnain, Hafsa Israr, **Muhammad Faris**, Rabika Kamal, Hafiza Syed Yusra Tirmizi. (2025). Evaluation of Machine Learning–Based Methods to Detect Bipolar Disorder in Individuals With Mental Health Conditions. *VFAST Transactions on Software Engineering*, 1–4 (HEC Y-category).
https://doi.org/10.21015/vtse.v13i3.2173

7. **Muhammad Faris**, Shahzaib Khalid, Muhammad Dilawar Khan, Mir Farooq Ali, Muhammad Mansoor Mughal, Tariq Javid. (2025). A SMOTE-Enhanced Hybrid VGG16–ResNet-50 Model for Automated Diabetic Retinopathy Detection. *Allied Medical Research Journal*, (HEC Y-category).
https://doi.org/10.59564/amrj/03.02/000

6. Tariq Javid, **Muhammad Faris,** Mir Farooq Ali, Muhammad Mansoor Mughal, Asad Saleem, Usama Zahoor. (2025). A Framework for AI-enabled Medical Device Design using Artificial Intelligent Agents. *Allied Medical Research Journal.* (HEC Y-category).
https://doi.org/10.59564/amrj/03.02/0010

5. **Muhammad Faris**, Tariq Javid, Khalid Mahmood, Danish Aziz, Mir Farooq Ali, Muhammad Mansoor Mughal. (2025). An Enhanced U-Net and CNN-based Tumor Edge Detection Technique in MR Images. *Allied Medical Research Journal.* (HEC Y-category).
https://doi.org/10.59564/amrj/03.01/019

4. Muhammad Mansoor Mughal, Tariq Javid, **Muhmmad Faris**. (2025). Investigation of brain computer interface-based exoskeleton in rehabilitation. *Pakistan Journal of Rehabilitation*, (HEC Y-category).
https://doi.org/10.36283/pjr.zu.14.2/012

3. Tariq Javid**, Muhammad Faris,** Arshad Aziz, Pervez Akhtar. (2022). Integrated representation for discrete Fourier and wavelet transforms using vector notation. *Mehran University Research Journal Of Engineering & Technology*, (HEC X-category).
https://doi.org/10.22581/muet1982.2203.18

2. **Muhammad Faris,** Tariq Javid, Majida Kazmi, Arshad Aziz. (2021). Segmented Multistage Reconstruction of Magnetic Resonance Images. *Journal of Independent Studies and Research Computing*, (HEC X-category IF=1.3).
https://doi.org/10.4316/AECE.2021.04012
1. Tariq Javid Ali, Pervez Akhtar, **Muhammad Faris**, Idris Mala, Syed Saood Zia. (2016). Implementation of Discrete Fourier Transform and Orthogonal Discrete Wavelet Transform in Python. *Journal of Independent Studies and Research Computing*,

Journal Accepted Articles

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Conference Publications

25. Syed Ibad Hasnain, Muhammad Faris, Crescenzo Pepe, Mir Farooq Ali, Silvia Maria Zanoli, **"AI-Driven Ensemble Approaches for Early Prediction of Cardiovascular Disease Risk",** In 2026, 27th International Carpathian Control Conference (ICCC), pp. 190-195. IEEE, 2026/1/6.
24. Laiba Azam, Muhammad Saim, Muhammad Faris, Mir Farooq Ali, Crescenzo Pepe, Silvia Maria Zanoli, **"Development of an AI-based Wearable HumanComputer Interface for Assistive Computing",** In 2026, 27th International Carpathian Control Conference (ICCC), pp. 66-71. IEEE, 2026/1/6.
23. M Asadullah Baig, Sarah Muzaffar, Muhammad Faris, Crescenzo Pepe, Mir Farooq Ali, "**Automated Contouring of Pelvic Computed Tomography Images for Radiation Therapy Systems using Deep Learning",** In 2026, 27th International Carpathian Control Conference (ICCC), pp. 72-77. IEEE, 2026/1/6.
22. **Muhammad Faris**, Ramsha Qayyum, Crescenzo Pepe, Mir Farooq Ali, Silvia Maria Zanoli, **"Multi-Class Skin Disease Detection Using Deep Learning Hybrid Method"** In 2025, 26th International Carpathian Control Conference (ICCC), pp. 1-5. IEEE, 2025/5/19.
21. **Muhammad Faris**, Fakiha Sohail, Crescenzo Pepe, Mir Farooq Ali, Silvia Maria Zanoli, **"Detection of Diabetic Retinopathy Using Deep Learning"** In 2025, 26th International Carpathian Control Conference (ICCC), pp. 1-4. IEEE, 2025/5/19.
20. **Muhammad Faris,** Muhammad Anees Jahangir, Muhammad Anas Naveed, Tariq Javid, Mir Farooq Ali, "**Detection of Pulse Transient Time through Single Lead ECG and PPG"** 40th All Pakistan IEEEP Students' Seminar 2025. February 20, 2025.
19. **Muhammad Faris,** Rameesha Khalid, Fakiha Sohail, Tariq Javid, Mir Farooq Ali, **"Detection of Retinopathy through Enhanced VGG-16 Model"** 40th All Pakistan IEEEP Students' Seminar 2025. Poster, February 20, 2025.
18. **Muhammad Faris,** Ramsha Qayyum, Fakiha Arshad, Khadija Razi, Tariq Javid, Mir Farooq Ali, **"Skin Disease Detection through Hybrid Deep Learning Method"** 40th All Pakistan IEEEP Students' Seminar 2025. **(Awarded Gold Category),** February 20, 2025
17. **Tariq Javid,** Muhammad Faris, Mir Farooq Ali, Muhammad Mansoor Mughal, Asad Saleem, Usama Zahoor "**A Framework for AI-enabled Medical Device Design using Artificial Intelligent Agents"** 3rd International Biomedical and Digital Health Conference (IBDC), April 24-25, 2024.
16. **Muhammad Faris**, Tariq Javid, Danish Aziz, Khalid Mehmood, Mir Farooq Ali, Muhammad Mansoor Mughal "**An** Enhanced U-Net and CNN based Tumor Edge Detection Technique to Improve Spatial Resolution in MR **Images"** 3rd International Biomedical and Digital Health Conference (IBDC), April 24-25, 2024.
15. Muhammad Mansoor Mughal, Tariq Javid, **Muhammad Faris "An Investigation of Brain Computer Interface based Exoskeletons in Rehabilitation"** 3rd International Biomedical and Digital Health Conference (IBDC), April 24-25, 2024.
14. Tariq Javid, **Muhammad Faris,** Sadia Ali, Fatima Irshad, Aliza Sohail, Arisha Shah, Omema Shakil. **"Design and Implementation of 4D Synthetic Human Pulse Generator Medical Device"** 1st International Conference on Emerging Trends in Biomedical Engineering, Science and Technology (ICET BEST) 2024, SHU, Karachi, February 28-29, 2024
13. Tariq Javid, **Muhammad Faris,** Hina Iftikhar, Tayyaba Khalid, Najam-ul-Saqib, Muhammad Bilal, Syeda Bazila. **"Artificial Intelligence based ECG for early detection of dysrhythmia"** 1st International Conference on Emerging Trends in Biomedical Engineering, Science and Technology (ICETBEST) 2024, SHU, Karachi, February 28-29, 2024

12. Tariq Javid**, Muhammad Faris,** Hina Iftikhar, Tayyaba Khalid, Maham Arshad**. "Artificial Intelligence based vital signs monitoring system**" 1st International Conference on Emerging Trends in Biomedical Engineering, Science and Technology (ICETBEST) 2024, SHU, Karachi, February 28-29, 2024
11. **Muhammad Faris,** Tariq Javid, Hina Iftikhar, Khalid Mahmood, Waleed Haider, Hareem Shakeel, Nameerah Nafees. **"Design of a Split AI System to Improve Spatial Resolution of Tumor Edge Detection in MR Images"** The 3rd Annual Biological Sciences Conference, AKU, Karachi, 15 November 2023 (Poster Presentation)
10. **Muhammad Faris**, T. Javid, SS H. Rizvi, and Arshad Aziz. **"Segmented Region-based Reconstruction of Magnetic Resonance Image."** In 2021 International Conference on Computer & Information Sciences (ICCOINS), pp. 68-73. IEEE, 2021.
9. Javid, Tariq, **Muhammad Faris**, Muhammad Danish Mujib, Tayyab Ahmed Ansari, Hina Iftikhar, Tayyaba Khalid, and Wardah Saadat. **"Cardiac Coronary Intervention Simulator."** In 2021 IEEE 18th International Conference on Smart Communities: Improving Quality of Life Using ICT, IoT and AI (HONET), pp. 66-70. IEEE, 2021.
8. Javid, Tariq, **Muhammad Faris,** Hira Beenish, and Muhammad Fahad. **"Cybersecurity and data privacy in the cloudlet for preliminary healthcare big data analytics."** In 2020 International Conference on computing and Information Technology (ICCIT-1441), pp. 1-4. IEEE, 2020.
7. **Faris, Muhammad,** Tariq Javid, Kashaf Fatima, Maliha Azhar, and Rohana Kamran. **"Detection of tumor region in MR image through the fusion of Dam construction and K-mean clustering algorithms."** In 2019 2nd International Conference on Computing, Mathematics and Engineering Technologies (iCoMET), pp. 1-16. IEEE, 2019.
6. Khalid, Eisha B., Afra Salman, Zeeshan Y. Khan, **Muhammad Faris,** Muhammad Danish Mujib, and Tayyab Ansari. **"Design and development of muscle monitoring system to visualize and monitor the activity of Bicep Brachii with varying load."** In 2017 International Conference on Innovations in Electrical Engineering and Computational Technologies (ICIEECT), pp. 1-4. IEEE, 2017.
5. **Faris, Muhammad**, Tayyab Ahmad Ansari, Tariq Javid, Abdul Haseeb, and Muhammad Danish Mujib. **"Towards design and development of noninvasive glucose monitoring system."** In 2017 International Conference on Innovations in Electrical Engineering and Computational Technologies (ICIEECT), IEEE, 2017.
4. **Javid**, Tariq, Pervez Akhtar, **Muhammad Faris**, Idris Mala, and Syed Saood Zia. **"Improving image contents using fuzzy rule-based contrast enhancement and histogram equalization: a comparative study."** IC2IS' 16
3. Faheem, Syeda Sheema, Qazi Bilal Ahmed, Aisha Nazir, and **M. Faris**, **"Design and development of fall detector and prevention system for Parkinson's patients."** In International Conference on Engineering and Emerging Technologies. 2016.
2. Ansari, Tayyab Ahmad, Ahmed Hassan, and **Muhammad Faris. "Development of Gesture based Wireless Appliances Control System for Hospital Ward."** Accepted Papers: 122. IMTIC'15
1. Sahar, Hooria, Shariqa Saher, **Muhammad Faris,** Tayyab Ahmad Ansari, and Khadija Ali Akber. **"Innovative Design of Intravenous Infusion System."** In Proc. of the 2nd Int. Conf. on Eng. and Emerg. Technol.: ICEET. 2015.

Funded Projects

NGIRI-2023-20530 **An Enhanced Deep Learning Based Tumor Edge Detection Technique to Improve Spatial and Temporal Resolution for MR Images**

NGIRI-2023-21587 **Designing a smart OT light equipped with high resolution cameras for shadow detection**

NGIRI-2020-4850 **Enhancement and Development of Low Cost Respiratory Monitoring and Controlling System**

NGIRI-2019-6328 **Detection of tumor region in MR image by using Raspberry Pi**

NGIRI-2019-5387 **Low cost Respiratory Monitoring and Controlling System**

Membership

- Member, Pakistan Engineering Council (Biomedical/325)
- Member, TECHNICAL COMMITTEE FOR "ELECTROMEDICAL EQUIPMENT (ESTC – 14)" PSQCA

Languages

- English: Professional working proficiency
- German: Basic proficiency (A1 Level)
- Urdu: Native proficiency

References

On demand